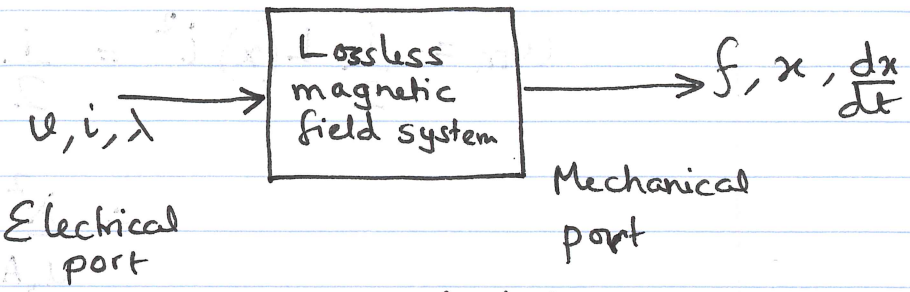
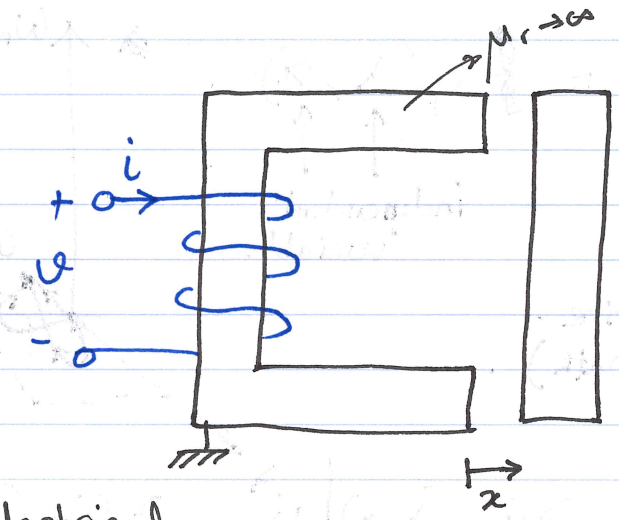
Lecture 18

Last time: *



Q. How do we calculate force? "Using virtual work"



Uniform cross-section area : A.

From electrical port perspective: $\lambda = L(x) \cdot i = f(x, i)$.

From conservation of energy:
W_m : Energy stored in the magnetic field:

$$\frac{dW_m}{dt} = vi - f_e \frac{dx}{dt}$$

or, $$\frac{dW_m}{dt} = i\frac{d\lambda}{dt} - f_e \frac{dx}{dt}$$

$$\therefore \quad \boxed{dW_m = i\,d\lambda - f_e\,dx} \qquad \text{——— (1)}$$

$\lambda = L(x) \cdot i$

In this example,

$$W_m = \frac{1}{2} L(x) i^2 = \frac{1}{2} \frac{\lambda^2}{L(x)} = \frac{1}{2} \frac{\lambda^2}{N^2} \cdot \frac{2x}{\mu_0 A}$$

$$= \frac{\lambda^2 x}{\mu_0 A N^2} = \frac{1}{2} \left(\frac{N^2 i^2 \mu_0 A}{2x} \right)$$

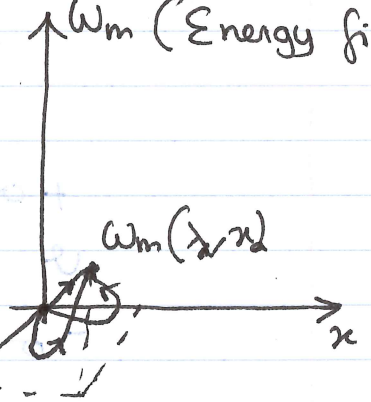
So, energy stored in the magnetic field in general,

$$W_m = \oint g(\lambda, x)$$

W_m (Energy field).

as this is lossless system, this is conservative. (independent of path).

independent variables



$$\therefore \quad \frac{dW_m}{dt} = \frac{\partial W_m(\lambda, x)}{\partial \lambda} \bigg|_{x=\text{const.}} \frac{d\lambda}{dt} + \frac{\partial W_m(\lambda, x)}{\partial x} \bigg|_{\lambda=\text{const.}} \frac{dx}{dt} \quad \text{——— (2)}$$

Relate (1) and (2).

$$f_e = - \frac{\partial W_m(\lambda, x)}{\partial x} \bigg|_{\lambda=\text{const.}}$$

In this example,

$$= - \frac{1}{2} \frac{\lambda^2 \cdot 2}{N^2 \mu_0 A} = - \frac{L(x) \cdot i^2}{N^2 \mu_0 A}$$

$$= -\left(\frac{N^2}{2x}\mu_0 A\right)^2 \frac{i^2}{N^2\mu_0 A} = -\frac{N^2\mu_0 A}{4x^2} i^2$$

$$\boxed{f_e = -\frac{N^2\mu_0 A}{4} \cdot \frac{1}{x^2} \cdot i^2}$$

↑ implies, force of electric origin is opposite to our convention of x.

$|f_e| \propto i^2$; $|f_e| \propto \frac{1}{x^2}$.

↑ independent of current polarity.

↑ infinite force at $x = 0$.

Word of caution : → How.

Step 1 : Calculate energy, ω_m as a function of λ and x.

Step 2 : $$f_e = -\left.\frac{\partial \omega_m}{\partial x}\right|_{\lambda=\text{const}}.$$

In essence, calculate energy stored in the field while no mech. work is done, calculate force when no energy is exchanged from the elec. port.

$$\omega_m = \left.\int_0^{\lambda} i\, d\lambda\right|_{x=\text{const}}.$$

$$= \int_0^{\lambda} \frac{\lambda'}{L(x)} d\lambda' = \frac{\lambda^2}{2L(x)}.$$

If you carefully look at

$$f_e = -\frac{\partial}{\partial x}\left\{\frac{\lambda^2}{2L(x)}\right\}\Bigg|_{\lambda=\text{const}} = -\lambda^2 \cdot \frac{\partial}{\partial x}\left\{\frac{1}{L(x)}\right\}.$$

In this force calculation approach λ is treated as constant and we have to invert inductance. λ is not directly measurable.
i is " ".

So, at the end we would like to express our force, as a function of x, and i.

So, you have to substitute back $\lambda = L(x) \cdot i$ after the derivative.

(λ, x) are independent variables.

Q. Is there an alternative approach of force computation?

~~We Know, $\lambda = f(x, i)$.~~

~~$\omega_m = f(x, i)$~~ ~~$i = i(t)$~~ $x = x(t)$.

~~$f_e = v\,i = i\,\frac{d\lambda(x,i)}{dt}$~~

~~$= i\left[\frac{\partial\lambda}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial\lambda}{\partial i}\cdot\frac{di}{dt}\right]$~~.

$$d(\lambda i) = \lambda\, di + i\, d\lambda.$$

$$\therefore \quad d(\omega_m) = d(\lambda i) - \lambda\, di - f_e\, dx.$$

$$\text{or} \quad d(\lambda i - \omega_m) = \lambda\, di + f_e\, dx.$$

co-energy $\leftarrow$ $d(\omega'_{co}) = \lambda\, di + f_e\, dx.$